

Practice Quiz: The Creative Agent

Part A: Multiple Choice

1. In Active Inference, an inventor's generative model includes: A) Only their technical knowledge and skills B) Beliefs about physics, user needs, markets, materials, and aesthetics — with varying precision C) A fixed blueprint of the invention they intend to build D) Only the information they have gathered in the current project
2. The "Methodical Engineer" inventor archetype is characterized by: A) Weak priors, rapid exploration, and tolerance for ambiguity B) Strong priors, high precision on technical feasibility, and systematic model refinement C) Broad but shallow knowledge across many unrelated domains D) Primary reliance on chance discoveries and serendipity
3. Why was Kodak engineer Steven Sasson able to invent the digital camera while senior photography experts missed the opportunity? A) He was more intelligent than the senior experts B) He had access to better equipment and funding C) His weaker priors about photography being necessarily chemical allowed him to perceive digital imaging as viable D) He was instructed by management to pursue digital technology
4. In multi-agent creativity, shared artifacts (prototypes, sketches) serve as: A) Final products ready for market B) Decorative elements that improve team morale C) Shared sensory evidence that multiple generative models can process and respond to D) Legal documents establishing intellectual property
5. The expected free energy that drives inventor motivation has two components: A) Cost and revenue B) Pragmatic value (achieving goals) and epistemic value (reducing uncertainty) C) Effort and reward D) Risk and return
6. Alexander Fleming's discovery of penicillin exemplifies the "accidental discoverer" archetype because: A) He had no relevant expertise and stumbled onto the discovery purely by luck B) He had strong domain expertise but also high precision on an unexpected observation that others dismissed C) He was systematically searching for

antibiotics through a methodical process D) He combined insights from multiple unrelated scientific fields

7. The optimal creative team balances: A) Identical generative models to ensure perfect communication B) Completely different models with no shared understanding C) Complementary models with enough shared framework to communicate but enough diversity to avoid groupthink D) A single leader's model with subordinates who execute without question

Part B: Short Answer / Analysis

1. The Wright brothers and James Dyson both produced breakthrough inventions through iterative prototyping, but their inventor archetypes differ. The Wright brothers tested many designs rapidly with minimal theoretical analysis (tinkerers), while Dyson built 5,127 prototypes guided by deep understanding of cyclonic physics (methodical engineer). Using Active Inference concepts, explain how both approaches can be effective despite their differences in prior strength and precision weighting. What type of problem is each approach best suited for?
2. A creative team consists of a mechanical engineer, a graphic designer, a marketing strategist, and a user researcher. Using the multi-agent Active Inference framework, describe how each team member's generative model contributes to the team's collective creative capacity. Identify a specific type of design decision where disagreement between team members would be most productive, and explain why using the concept of complementary precision weightings.
3. You are an inventor who identifies strongly as a Methodical Engineer (strong priors, high precision on technical feasibility, systematic approach). Your current project requires a radically novel solution — something outside the domain of your existing expertise. Using Active Inference concepts, describe three specific strategies for temporarily shifting your agent configuration to be more exploratory. For each strategy, explain what Active Inference parameter it modifies and how.